

Cognitive Decline Prediction based on Social Determinants: Model Report

Carlos Huertas

Team: NxGTR

Email: ing.carloshuertas@gmail.com

Abstract—In this work we address the challenge to identify early signs of cognitive decline based on social determinants seeking to improve early detection for Alzheimer’s disease and related dementias. The strength of our work is validated by a DrivenData ML competition with over 170 individual participant, including public and private organizations. Our design approach relies on minimum complexity but maximum performance. We provide guidelines on how different modeling design decisions impact performance, starting with a minimalist solution that still achieves high-performance, all the way to state-of-the-art scores from our final solution based on Gradient Boosting Decisions Trees. We introduce the concept of cognitive decline acceleration, a different way to formulate the problem of decline, this proves to be key to achieve a 3rd best prediction score in the competition. Our model exhibits low bias for most protected classes, but further work is required for older adults. We finally review potential next steps to improve this solution and close with outstanding questions for subsequent research.

I. INTRODUCTION

Alzheimer disease and related dementias (ADRDs) refer to a group of common neurodegenerative illnesses that can lead to short-term memory loss, paranoia, and delusional ideas [1]. By 2060, it is estimated these conditions will affect 13 million Americans over 65 years old [2] which will represents around 13.7% of older adults [3]. As populations get older, there is increased interest in understanding these diseases to enhance patients’ life. Alzheimer’s Disease (AD) has no treatment and can last for years or end-of-life and early detection provides a much better opportunity for recovery [4]. Prior work towards ADRD early detection can require very sophisticated data collection [5], a new approach leveraging Social Determinants [6] provide alternative methods using much easily accessible data since they represent non-medical attributes like employment, income, and age [7]. In particular, Social determinants of health (SDHs) can be categorized in upstream & downstream [8], while the latter are more focus on health-related attitudes, the former are fundamental factors that lead health effects even if not directly correlated, e.g. education. In this work, we review data from the Mexican Health and Aging Study (MHAS) which includes information about demographics, economic circumstances, migration, physical limitations, self-reported health, and lifestyle behaviors of adults over 50 years old. The dataset includes in-person interviews from surveys during 2003 & 2012, and further composite score reflects cognitive functions over subsequent years 2016 & 2021. Composite scores are calculated based on in-depth

cognitive assessments that were administered in person as part of the MHAS Cognitive Aging Ancillary Study (Mex-Cog). Our research objective is to determine if SDH’s are useful early predictors of future cognitive decline. In order to validate the value proposition of our approach, an online competition brings researches around the world under the DrivenData platform to evaluate performance of each proposal.

II. METHODOLOGY AND MODEL PERFORMANCE

All experiments rely exclusively on the data provided by MHAS in collaboration with the National Institute of Aging (NIA) for the DrivenData competition “*PREPARE Challenge - Phase 2: Model Arena (Social Determinants Track)*” (see <https://www.drivendata.org/competitions/300/>).

Our first modeling decision comes from simplifying the multi-year records and allowing the model on its own to determine the appropriate usage of *year* signal. Once the train set has been expanded to account for multi-records (e.g. same candidate with Composite Score (CS) in both 2016 & 2021), we have a total of 4343 samples from 3276 unique patients. Since our task involves building a solution capable to estimating the future CS, a regression approach is needed. One fundamental problem with these models is lack of interpretability from their metrics, e.g. Root Mean Squared Error (RMSE) has no particular meaning without context [9], unlike accuracy which is bounded (0 to 100%), RMSE has no upper bound.

In order to give a meaningful human-readable value to RMSE, we can see it as a relative metric, e.g. how the value decreases (smaller is better) with respect to a baseline. This allows us to quantify the actual real-life improvement of a model. In Table I we provide trivial baselines and observed performance for the *train-set*. Worth notice is the fact that *Education* provides a strong baseline, this is a simple aggregate of observed scores that achieves a $\approx 23\%$ improvement over the most naive approach, calculated as $improvement = (46.9/60.9) - 1$. Other features like *occupation type* and *age* provide some marginal gains only. The final of our naive baselines involve zero-adjustments, using data as-is, a single decision tree (DT) (*leaves=16*) barely outperforms the trivial single-feature *education benchmark*. This suggests that the vast majority of the predictive power comes from *education* and the problem exhibits a strong deep non-linear pattern. Additionally, by doing a K-Fold validation

TABLE I
MHAS: DRIVENDATA COMPETITION NAIVE BASELINES FOR TRAIN-SET

Baseline	Group Variable	RMSE	Improvement over Global Mean
Composite Score Global Mean	n/a	60.9025 $\pm$ (0.0)	n/a
Longest Occupation Type Conditional Mean	rjlocc_m_12	56.1575 $\pm$ (0.0)	7.79%
Age Conditional Mean	age_12	55.4768 $\pm$ (0.0)	8.90%
Education Conditional Mean	edu_gru_12	46.9058 $\pm$ (0.0)	22.98%
Single(1) Decision Tree (KFold=5)	Entire set (184)	45.71 $\pm$ (1.97)	24.94%

we estimate standard deviation (SD) in the ≈ 2 range, this serves as guideline to understand the stability of results and how statistically significant are deltas between all available solutions.

Since Gradient Boosting Decision Tree (GBDT) is the state-of-the-art for tabular data [10] and it has shown promising results in prior research for Alzheimer’s Disease Prediction [1], we dedicated our efforts in this direction only. Our modeling uses the LightGBM [11] algorithm exclusively for all further purpose. This decision is inspired by simplicity and explainability as top priorities since its trivial to improve further by means of stacking or ensembling several models.

A. Default LightGBM

The most trivial solution involves using a LightGBM model with the data as-is, no pre-processing of any kind (e.g. not even missing value imputation). Categorical features are also not-processed at all, being left as *category* for the algorithm to decide. This is evaluated over a KFold=5 strategy and achieves a $RMSE = 41.12 \pm (1.32)$. The results indicate how much powerful this technique is over a single decision tree, even though its executed with all default parameters. For context, this solution would have outperformed $\approx 28\%$ of leaderboard solutions. Early experiments suggest Neural Network (NN) underperformed by around 5% which could provide insights about this unusual good performance with such a naive baseline.

B. Tuned LightGBM

Besides performance improvements, parameter tuning can help the model to generalize beyond a ML-metric (e.g. RMSE), by enforcing real-life constraints, we can force the model to be conservative, in particular, we want to ensure no prediction is ever based on single-observations, since a DT can produce leaves of $size = 1$, we add a constraint with $min_data_in_leaf = 5$. This is a desirable model behavior regardless of metrics as it ensures predictions can always have precedence of known-cases, since all scores have at least 5 patients to rely on. In addition to this, further penalties in the form of L1 and L2 regularization is included.

The full set of parameters can be found in Table II. The key learnings in parameter tuning was a strong focus in regularization due to the small data size. This model is now able to achieve $RMSE = 40.48 \pm (1.39)$, which already approaches around half the competition leaderboard outperforming $\approx 44\%$.

TABLE II
LIGHTGBM MHAS SPEED MODEL PARAMETERS

Parameter	Default Value	Tuned Value
learning_rate	0.1	0.01
num_leaves	31	32
min_data_in_leaf	20	5
feature_fraction	1.0	0.6
bagging_fraction	1.0	0.6
lambda_l1	0.0	3
lambda_l2	0.0	6
path_smooth	0	4

C. Feature Encoding

Unfortunately, the data in this competition is too small for the feature size, although studies have found its complicated to generalize the optimal sample-size (SS) to feature-size (FS) ratio [12], a SS far exceeding FS is preferable. In this problem we have 184 features & 3276 unique patients, for a ratio of 1:17.8. A ratio of over 1:50 would be preferable. To mitigate this issue we apply feature target encoding. Our transformation replaces all features values with their conditional observed response (*composite_score*) mean. This works very well due to the small data size and model inability to properly generalize on its own. Worth notice however is this technique requires feature-value regularization to eliminate single-value-observations as such samples observed mean is the sample itself leading to straight overfit.

After encoding, our new model is now achieving $RMSE = 39.00 \pm (1.37)$, this change alone provides over $2x$ the improvement observed during parameter tuning. This a simple, but strong solution already approaching Top #10 results in the competition leaderboard.

D. Feature Engineering

As described in Section I, our modeling approach duplicates samples and allows the model to learn how the same representation (e.g. the same patient) should lead to different outcomes for each year. To aid this, a feature indicates to the model if samples are found for both 2016 & 2021 evaluations. Our intuition of the power behind this feature comes from bias exhibited by patients willing to continue their study over the years, i.e., the fact patients have attended a previous session (e.g. 4 years ago) and are present now for a subsequent evaluation shows care for their health.

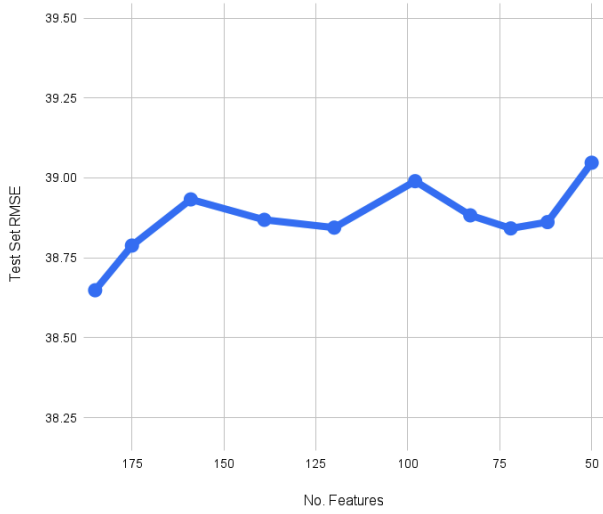


Fig. 1. Feature Size vs Performance

Thanks to this count feature, our best model now achieves $RMSE = 38.69 \pm (1.28)$, this also marginally reduces SD, showing smaller fluctuations. This continues to be a single-model capable of reaching around Top #5 in the competition.

E. Feature Selection

Although Feature Selection did not provide strong performance improvements, it is a desirable step for real-life production environment due to its impact on data-collection. In this ML competition data was already prepared, but in a real-world setting vast efforts are spent in collecting & cleaning data, we studied the effect of variable reduction with respect to performance. It is possible to remove $\approx 70\%$ of variables, and build a 50-features model that still scores a Top #10 solution. The effect in performance is shown in Figure 1.

F. Additional Speed Model

Although not recommended for real-life, in the context this competition we opted for developing a secondary speed-model using the CatBoost [13] algorithm. This solution uses the exact same data, pre-processing and feature engineering as described earlier in this section. The only change is a retrain using a different algorithm, the performance is almost identical but it provides partially uncorrelated predictions that serve ensembling if required.

G. Basic Modeling Summary

We have developed what is known as a point-prediction estimate, a regression model which output represents a single-value to approximate the CS given a set of SDH for two different time periods. The model is able to output scores per known-years (2016 & 2021). We call this the “*Speed Model*” which estimates a rate of change for “*position*”, in this case a CS, but unable to determine how fast the change would happen beyond the observed timeframe, e.g. 2026, to achieve this we can find the “*Acceleration*”. This will be further discussed in

Section III. Our “*Speed Model*” performance details can be found in Table III.

III. INNOVATION AND INSIGHTS

A. Innovation

Our approach introduces two innovations to this problem, first a data-driven feature cleaning improves generalization without the need to drop noisy features. Second, we introduce the concept of “*Acceleration*” to further model the rate of change in “*speed*” with respect to “*time*”, this is, two very similar patients might lead to the *same* 2016 composite score, yet if they arrive at such score with different “*Acceleration*” their 2021 decline can be inconsistent, in fact, it can move in the opposite direction.

1) *Feature Cleaning*: As described in Section II-C, sample size is too small for strong generalization. In order to improve this, we apply a regularization technique directly into the data-itself, essentially as a model-agnostic constraint. The approach consists on enforcing a *minimum-value-count-constraint* (MVCC); this requires any value observed to be present in both out-of-distribution test & at least 2-times in train. For example, the variable *earnings_03* representing earnings in employment during 2013 has 142 unique-values with mean(std) of $\$13,113(\pm\$33,769)$, where the SD far exceeds the mean, this is undesirable behavior for such small dataset. After applying MVCC, unique values decrease to 67 with a range of $\$7,372(\pm\$18,557)$. MVCC does *not* lead to improvements in random KFold validation, but improves generalization for real-life performance. This is mostly caused by KFold measuring overly confident results (overfitting). Thanks to MVCC, local validation strongly matches the competition private leaderboard score which remained hidden during the entire development.

2) *Acceleration Model*: In Section II we provide the base “*Speed Model*” design; which is able to determine the rate of change in *position* for a CS spectrum. In general, most of the performance comes from such model, and represents a strong single-model solution for Top #5. If extra performance is needed, a secondary model, known as *Acceleration model* gives additional boost for multi-year instances. To achieve this, we formulate a derivative for this problem as defined in Equation 1 & 2 respectively for speed & acceleration.

$$speed_decline(t) = \frac{d}{dt} (composite_score(t)) \quad (1)$$

$$acceleration_decline(t) = \frac{d}{dt} (speed_decline(t)) \quad (2)$$

The “*Acceleration Model*” is trained to learn the rate of change observed for patients between 2016 & 2021. This prediction can further be applied to the “*Speed Model*” and refine predictions. An example of this improvement can be observed for *UID_{bqdf}* which shows a *speed-model* score of 146 and 135 for 2016 & 2021 respectively, or, as seen by *acceleration-model*, a -11 deceleration.

TABLE III
NIA DRIVEN DATA COMPETITION: SPEED MODEL PERFORMANCE (KFOLD5 VALIDATION)

Model Setup (LightGBM)	Details	RMSE	Improvement over Global Mean	Improvement over Single DT
Default Params + Raw Data	Section II-A	41.12 $\pm$ (1.32)	32.48%	10.04%
Tuned Params + Raw Data	Section II-B	40.48 $\pm$ (1.39)	33.53%	11.44%
Tuned Params + Encoded Data	Section II-C	39.00 $\pm$ (1.37)	35.96%	14.67%
Tuned Params + Feature Eng	Section II-D	38.69 $\pm$ (1.28)	36.47%	15.35%

TABLE IV
FINAL SOLUTION FEATURE IMPORTANCE SUMMARY

Feature Name	Description	Speed Imp.	Acc Imp.	Global Imp.
edu_gru_12	Binned education level (2012)	1.000	1.000	1.000
edu_gru_03	Binned education level (2003)	0.306	0.282	0.294
j11_12	Floor material of residence (2012)	0.044	0.443	0.244
rjob_end_03	Year that their last job ended (2003)	0.016	0.385	0.200
hincome_12	Household income (2012)	0.072	0.267	0.170
rsocact_m_12	How often they have social activities (2012)	0.051	0.229	0.140
rrfcntx_m_12	How often they see friends and relatives (2012)	0.058	0.222	0.140
rjlocc_m_12	Category of their longest occupation (2012)	0.139	0.116	0.128
urban_12	Locality size. Either rural or urban (2012)	0.015	0.231	0.123
age_12	Binned age group (2012)	0.165	0.078	0.122

However, the true deceleration as reported in the ground-truth labels is of -76, a large change from 162 points in 2016 to only 86 in 2021. The acceleration model improves this by suggesting a -28 deceleration, over 200% improvement than using speed-model alone. This technique can boost performance further by ≈ -0.05 RMSE which roughly improves one top position up in the leaderboard. This might appear small in magnitude, but as competitors get close in scores, this delta is similar to the difference between Top #12 and Top #17 teams.

B. Insights

Model insights are mostly derived from key feature importance understanding. A detailed table showing normalized (0 to 1) scores per model is shown in Table IV. Main insights are described as follows:

1) *60% of features are not relevant:* As reviewed in Section II-E and Figure 1; several features provide little-to-zero value for this task. Lack of standardization in data collection is an issue, for instance *adl_dress_03*, which defines difficulties to get dressed, show promising value for 2012, but not for 2003 data. This however, does not mean that all data during 2003 is not useful, out of 44 features removed, 20 of them (45%) belong to recent timeframe (2012). Although models can incur a ≈ 0.2 RMSE penalty for removing so many features, this is negligible compared to real-world benefits of such leaner model. Efforts can be better geared toward collecting other type of signals leading to potential improvements and recover the deficit.

2) *Education Level is unproportionally strong:* As it can be seen in Table I & IV, *Education Level* alone, as provided by

features *edu_gru_03* & *edu_gru_12* provides most of model performance, with 92% of predictive power. Given this, it would be of vital importance to expand information collection on this field. Efforts should be spent digging further a more granular understanding of this signal.

3) *Income 6x more important for Acceleration:* While income, e.g. *hincome_12* appears with very low predictive power for *speed-model*, its a top importance feature for *acceleration-model*. Another related signal regards residence floor materials (e.g. *j11_12*), this is strongly correlated with income and exhibits same behavior. Normalized feature importance shows 6.12x more value for those signals in favor of acceleration. This important insight reveals that Income is not exactly powerful to determine where the patient will land in the spectrum in a given year, but very important on how things will progress further, even for years (e.g. 2026) outside the competition data.

4) *Social Activities as Top Actionable Item:* In the real-world, not everything is actionable, e.g. one might never be able to change area of longest occupation, even though this data can be used as predictor. On the other hand, overall Top #6 & #7 features shows promising results to demonstrate that social activities play an important role in cognitive health. Both signals (*rsocact_m_12* & *rrfcntx_m_12*) measure the frequency on engaging in some form of social activity, including staying in touch of friends and relatives.

IV. BIAS EXPLORATION AND MITIGATION

Since ML are function approximators that learn from data, it is the data-itself that introduces bias [14], it is of great importance to evaluate how the model generates predictions

TABLE V
MODEL BIAS EXPLORATION RESULTS

Protected Class	Feature	Group	MAPE
Age	age_12	-59	19.87%
		+60	34.74%
		Bias	1.74x
Religion	rrelgwk_12	No	29.28%
		Yes	29.62%
		Bias	1.01x
Gender	ragender	Woman	30.56%
		Man	32.28%
		Bias	1.05x
Sexual Orientation	sgender_12	Heterosexual	31.24%
		Homosexual	1-sample
		Bias	No Data

for protected groups, and one of the main modern techniques involve Disparate Impact [15]. In terms of bias mitigation, adjusting the data to more fairly represent each population is desirable, this can involve re-weighting and sensitive fields removal [16]. In this work we first introduce a measurement of bias based on error-rate and provide mitigation proposals following a similar framework as AIF360 [17]. Using as reference protected classes from the Public Health Service Act [18] we identify this dataset showcase: *Age*, *Religion*, *Gender* & *Sexual Orientation*. We evaluate disparity of model performance across classes using *Mean Absolute Percentage Error (MAPE)* since it provides a normalized error per group, unlike RMSE which introduce bias as groups can have different means. This is very important as one could easily hide the bias and report apparently fair models by using the wrong metric. For instance, if we were to use differences based on RMSE, the age-group bias would suddenly disappear, yet on closer inspection it's not an accurate representation. Results are shown in Table V.

The model behaves reasonable with very low bias in all categories with the exception of *Age*, which shows a high bias of 1.74x (34.74/19.87) where older population shows a higher relative error-rate. As metioned before, this can only be observed with MAPE.

In terms of mitigation, our solution already addresses part of it with strong regularization mechanisms as shown in Table II, this help prevent over indexing in particular features and bias predictions. We studied further the remaining observed bias with *age*. One actionable item is to remove direct features like *age_12* & *age_03*, these signals however are of key importance and their removal brings a performance penalty of $1.55RMSE$ and does not reduce any bias since the model is able to reconstruct age via feature correlations. A key root cause of performance disparity comes from data bias where older individuals are consistently inversely correlated with education, with 54% of patients over 60 with not completed elementary school compared to 32% in under 60 years old. Naturally the same trend is observed at higher education, where 20% of young patients have high-school or higher, but

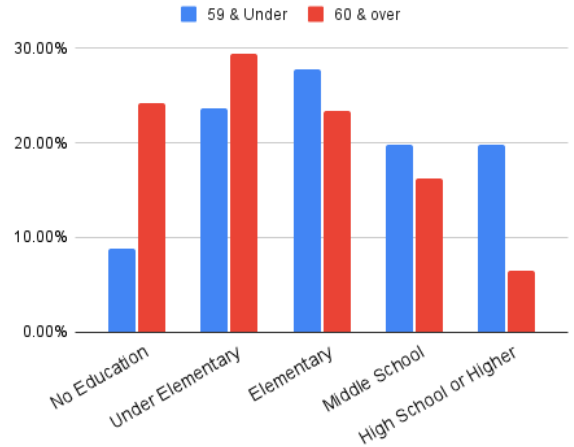


Fig. 2. Education Distribution per Age Group

only 6% of older adults do. Since education (*edu_gru_12*) is the strongest predictor, the bias is expected. Ideally a more inclusive data collection process can help mitigate this bias but under the existing competition setting improvements to this bias bring too harsh performance penalties. Full detail of education distribution can be observed in Figure 2.

V. EXPLAINABILITY

Full details will be provided in a dedicated report per competition reporting guidelines.

VI. GENERALIZABILITY

Our solution allows for practical generalization across different sets due to its simplicity. For reference, we provide a proof-of-concept code solution under 10-lines of code which can achieve $RMSE = 39.31$, enough for a Top #17 in the competition, see Algorithm 1.

Even though it is possible to increase performance by mass ensembling & stacking, e.g. by using techniques like AutoGluon [19] which trains over 130 models, it is not worth the practical implications such as training time, inference latency and maintenance costs. Our single (1x) LightGBM model shows superior stable results, with internal cross-validation of $RMSE = 38.69$ (see Table III) matching almost exactly the final score of $RMSE = 38.65$ as seen in Table VI.

Our *acceleration-model* proved to generalize well and is able to correct mistakes from both *speed-models*. This approach is responsible for moving our solution to 3rd place, the $-0.05RMSE$ boost in performance is consistent across different models which we believe can improve other competitors solutions as well.

No pre-trained data of any kind has been used for this work; the entire solution depends uniquely on the provided data for this competition. Increased generalization to other domains is possible since no additional dependencies are introduced.

The simplicity of this works also allows for model building in regular consumer PCs, with processing completing end-

Algorithm 1 Simple End-to-End Train Pipeline

```
import pandas as pd; import numpy as np; import lightgbm as lgb;
tr_lbl = pd.read_csv('train_labels.csv')
tr = pd.read_csv('train_features.csv').merge(tr_lbl, how='left', on='uid')
for eF in tr.columns[1:-2]:
    tr[eF] = tr[eF].map(tr.groupby(eF)['composite_score'].median().to_dict())
gt = tr['composite_score']
tr.drop(['composite_score', 'uid'], axis=1, inplace=True)
params = { 'objective': 'rmse', 'num_leaves': 2*5, 'learning_rate': 0.01,
           'min_data_in_leaf': 6, 'feature_fraction': 0.5, 'lambda_l1': 3.0,
           'bagging_fraction': 0.5, 'lambda_l2': 7.0, 'bagging_freq': 1,
           'n_estimators': 790, 'path_smooth': 4}
model = lgb.train(params, lgb.Dataset(tr, label=gt))
```

TABLE VI
DRIVENDATA COMPETITION: LEADERBOARD SUBMISSION RESULTS

Solution	Score	Position
Simple Code: Algorithm 1	39.3123	17th
1x LightGBM Speed Model	38.6486	5th
1x CatBoost Speed Model	38.6946	5th
Dual Speed Model	38.5301	4th
Dual Speed + Acc Model	38.4815	3rd

to-end in a few seconds with simple CPU and without any requirements for Graphic Processing Units (GPU).

VII. FUTURE DIRECTION

The data in this competition seems to be a post-processed version of the original set collected by Mexican Health and Aging Study (MHAS), since improvements in data tend to bring much higher gains than investment in modeling [20], our first action would involve the study of any raw available sets of data and try to improve existing processing functions, in particular signals related to education could provide performance boost with different data management, including, but not limited to embedding generations.

Another direction we would pursue involves the re design of the data collection process. After our findings with feature selection (see Figure 1), it seems a lot of the data does not contribute for the task. It is important to understand how the signals are improving other downstream tasks, and if not, efforts can be better spent collecting new signals. Working backwards from existing error-rates can guide a data-driven collection methodology.

Finally, we can explore more complicated modeling techniques, but based on existing state-of-the-art results from the competition it will require extensive analysis to justify the performance-complexity tradeoff. We need to understand how the model can be used by health practitioners and how sensitive their actions are to small variations in prediction scores.

VIII. CONCLUSIONS

A ML competition is a great way to remove biases and conflict of interest regarding what a best solution is. Results have proven that social determinants provide predicting power to identify early signs of Alzheimer’s disease and Alzheimer’s disease related dementias (AD/ADRD). What this paper does not address is the usefulness of these models in real-life settings. Even though our method provides state-of-the-art results, error rates from all competitors exceed 38 points in the composite score scale, a naive solution relaying only on mean observed values per education level is less than 9 points away ($RMSE = 46.9$), this opens serious questions on how applications based on this performance could actually have an impact in people. Further partnership with medical practitioners is required to validate not only performance but how actionable these predictions are. This work also only explored a maximum of 9-year delta, since early prediction is critical, further work needs to explore how larger time frames balance prediction power and actionable items. ML models with very high performance but simple design continues to be a challenge, automatic ML solutions are still no match for human ingenuity in some challenges.

REFERENCES

- [1] C. Kavitha, Vinodhini Mani, S. R. Srividhya, Osamah Ibrahim Khalaf, and Carlos Andrés Tavera Romero, “Early-stage alzheimer’s disease prediction using machine learning models,” *Frontiers in Public Health*, vol. 10, 2022.
- [2] Kevin A Matthews, Wei Xu, Anne H Gaglioti, James B Holt, Janet B Croft, Dominic Mack, and Lisa C McGuire, “Racial and ethnic estimates of alzheimer’s disease and related dementias in the united states (2015-2060) in adults aged ≥ 65 years,” *Alzheimers Dement*, vol. 15, no. 1, pp. 17–24, Sept. 2018.
- [3] Census.gov, “U.s. census bureau, 2017 national population projections,” 2017.
- [4] R. Sivakani and Gufran Ahmad Ansari, “Machine learning framework for implementing alzheimer’s disease,” in *2020 International Conference on Communication and Signal Processing (ICCSPP)*, 2020, pp. 0588–0592.
- [5] Protima Khan, Md. Fazlul Kader, S. M. Riazul Islam, Aisha B. Rahman, Md. Shahriar Kamal, Masbah Uddin Toha, and Kyung-Sup Kwak, “Machine learning and deep learning approaches for brain disease

diagnosis: Principles and recent advances,” *IEEE Access*, vol. 9, pp. 37622–37655, 2021.

- [6] Muniza Anum Majoka and Corbett Schimming, “Effect of social determinants of health on cognition and risk of alzheimer disease and related dementias,” *Clinical Therapeutics*, vol. 43, no. 6, pp. 922–929, 2024/11/26 2021.
- [7] World Health Organization, “Social determinants of health,” .
- [8] Paula Braveman, Susan Egerter, and David R. Williams, “The social determinants of health: Coming of age,” *Annual Review of Public Health*, vol. 32, no. Volume 32, 2011, pp. 381–398, 2011.
- [9] T. Chai and R. R. Draxler, “Root mean square error (RMSE) or mean absolute error (MAE)? - Arguments against avoiding RMSE in the literature,” *Geoscientific Model Development*, vol. 7, no. 3, pp. 1247–1250, June 2014.
- [10] Vadim Borisov, Tobias Leemann, Kathrin Seßler, Johannes Haug, Martin Pawelczyk, and Gjergji Kasneci, “Deep neural networks and tabular data: A survey,” *IEEE Transactions on Neural Networks and Learning Systems*, p. 1–21, 2024.
- [11] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu, “Lightgbm: A highly efficient gradient boosting decision tree,” in *Advances in Neural Information Processing Systems*, I. Guyon, U. Von Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, and R. Garnett, Eds. 2017, vol. 30, Curran Associates, Inc.
- [12] Jianping Hua, Zixiang Xiong, James Lowey, Edward Suh, and Edward R Dougherty, “Optimal number of features as a function of sample size for various classification rules,” *Bioinformatics*, vol. 21, no. 8, pp. 1509–1515, 2004.
- [13] Anna Veronika Dorogush, Andrey Gulin, Gleb Gusev, Nikita Kazeev, Liudmila Ostroumova Prokhorenkova, and Aleksandr Vorobev, “Fighting biases with dynamic boosting,” *CoRR*, vol. abs/1706.09516, 2017.
- [14] Ninareh Mehrabi, Fred Morstatter, Nripsuta Saxena, Kristina Lerman, and Aram Galstyan, “A survey on bias and fairness in machine learning,” *ACM Comput. Surv.*, vol. 54, no. 6, July 2021.
- [15] Utsab Khakurel, Ghada Abdelmoumin, Aakriti Bajracharya, and Danda B. Rawat, “Exploring bias and fairness in artificial intelligence and machine learning algorithms,” in *Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications IV*, Tien Pham, Latasha Solomon, and Myron E. Hohil, Eds., June 2022, vol. 12113 of *Society of Photo-Optical Instrumentation Engineers (SPIE) Conference Series*, p. 1211324.
- [16] Faisal Kamiran and Toon Calders, “Data preprocessing techniques for classification without discrimination,” *Knowl. Inf. Syst.*, vol. 33, no. 1, pp. 1–33, Oct. 2012.
- [17] Rachel K. E. Bellamy, Kuntal Dey, Michael Hind, Samuel C. Hoffman, Stephanie Houde, Kalapriya Kannan, Pranay Lohia, Jacquelyn Martino, Sameep Mehta, Aleksandra Mojsilovic, Seema Nagar, Karthikeyan Nate-san Ramamurthy, John Richards, Diptikalyan Saha, Prasanna Sattigeri, Moninder Singh, Kush R. Varshney, and Yunfeng Zhang, “AI Fairness 360: An extensible toolkit for detecting, understanding, and mitigating unwanted algorithmic bias,” Oct. 2018.
- [18] Public Health Service Act, “Title 42: Chapter 6a - section 1908,” .
- [19] Nick Erickson, Jonas Mueller, Alexander Shirkov, Hang Zhang, Pedro Larroy, Mu Li, and Alexander Smola, “Autogluon-tabular: Robust and accurate automl for structured data,” *arXiv preprint arXiv:2003.06505*, 2020.
- [20] Carlos Huertas and Qi Zhao, “ On The Irrelevance of Machine Learning Algorithms and the Importance of Relativity ,” in *2023 IEEE International Conference on Multimedia and Expo Workshops (ICMEW)*, Los Alamitos, CA, USA, July 2023, pp. 16–21, IEEE Computer Society.